

The Market for Lemons and the Regulator's Signalling Problem

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Abstract

The Market for Lemons is a classic model of asymmetric information first studied by Nobel Prize economist George Akerlof. It shows that information asymmetry between the seller and buyer may result in market collapse or some sellers leaving the market. “Lemons” in the used car market are cars of poor quality. The information asymmetry present is that the buyer is uncertain of the cars’ true quality. I first offer a simple baseline model that illustrates the market collapse, and then examine what happens when regulation, ie. a DMV is introduced to reveal (signal) the true car quality to the buyer. The effect on the market varies based on the assumptions about the regulator. The central focus is on the DMV’s signal structure, which can have interesting effects on the market and the information asymmetry. I show that surprisingly, when the DMV actually decreases the quality of their signal in a well constructed way, it can substantially increase their profit. On the other hand, this negatively effects overall welfare.

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1 Baseline Model

In this article, we will review the famous Market for Lemons by Akerlof. It is a simple model that shows how information asymmetry between buyer and seller can have very interesting effects on a market. Then we will introduce a certain third party and study the implications of information design.

“Lemons” in a market are essentially bad goods of low quality. Akerlof showed that information asymmetries in the market can cause the quality of goods to degrade until only “lemons” are left.

We use a car market for our setting as in Akerlof’s model. The information asymmetry here is that the quality of the car is known to the sellers (car dealers) but not to buyers. We can represent the quality of a used car by a hidden type $\theta \in [0, 1]$, where $\theta = 0$ means a busted car and $\theta = 1$ means in new condition.

Payoffs for the buyer and seller are given by:

Market for lemons

	no sale	sale
seller	$\pi\theta$	p
buyer	0	$\theta - p$

where $\pi < 1$ indicates that the seller is somewhat tired of the car and it has less utility for them (especially the longer they’re unable to sell the car), though π is not too small, so we assume $\pi > \frac{1}{2}$. And p is the price of the car determined endogenously by the market.

Now each seller has a type represented by their $\theta \in [0, 1]$. Suppose all sellers are on the market. Then we note that the buyer accepts p only if $p \leq E(\theta) = \frac{1}{2}$ under uniform prior. But then given $p = \frac{1}{2}$, seller gets net gain of $p - \pi\theta = \frac{1}{2} - \pi\theta$, so she’ll only sell if $\theta \leq \frac{1}{2\pi} < 1$.

Now since only the sellers with $\theta \leq \frac{1}{2\pi}$ will sell, the buyer will only buy if $p \leq E(\theta \mid \theta \leq \frac{1}{2\pi}) = \frac{1}{4\pi}$. But by similar reasoning then the seller will only sell if $\frac{1}{4\pi} - \pi\theta \geq 0$, or $\theta \leq \frac{1}{4\pi^2} < \frac{1}{2\pi}$.

This reasoning can easily be repeated recursively, and it follows by induction at the n -th step that the only car dealers left willing to sell are $\theta \leq \frac{1}{(2\pi)^n}$ and buyers will only buy if

$\theta \leq \frac{1}{2(2\pi)^n}$ but clearly as $n \rightarrow \infty$ the $\frac{1}{(2\pi)^n} \rightarrow 0$, so in fact only $\theta = 0$, the lemons will sell. Do we have a market anymore? Apparently not.

This kind of reasoning is called unraveling to the bottom, or retrograde analysis or working backwards. When this was first proposed, it had profound implications, because it showed that under reasonable assumptions and information asymmetry, a market could suffer market failure.

What is a proposed solution? There are indeed market solutions on both supply and demand side. On the supply side, sellers may provide a warranty to assuage buyers' concerns about quality uncertainty, but this may be costly for sellers with low quality cars. The sellers may also increase the proportion of higher quality cars, so the buyer's prior belief of buying a good car is higher. However this may not be so easy to achieve for a particular seller type. On the demand side, buyers may seek third party quality specialists to examine the cars, thereby removing the information asymmetry. We consider a third alternative, a regulation based solution to create a different kind of incentive in the market.

Let us assume the DMV, a regulator, will assist the buyer in determining the true value of the seller's car quality, thereby mitigating the effects of the information asymmetry.

Suppose the DMV offers certificates that disclose quality for free. (We will re-evaluate this assumption later.) Each seller then chooses whether to disclose θ or not. Thus, the buyer is willing to pay for

- $w = \theta$ if θ is disclosed, and
- $w = E(\theta \mid \text{undisclosed}) = \hat{\theta}$ if θ is not disclosed.

However, note that given $\hat{\theta}$, the seller types $\theta \geq \hat{\theta}$ will disclose, and sellers with $\theta < \hat{\theta}$ are better off not disclosing. But this means that $\hat{\theta} = E(\theta \mid \text{undisclosed}) = E(\theta \mid \theta < \hat{\theta}) = \frac{\hat{\theta}}{2}$, which solves $\hat{\theta} = 0$. Thus, only the lemons won't disclose.

We can easily understand this substantively with a similar recursive argument. Suppose some sellers θ disclose and some don't disclose. Consider a nondisclosing seller with θ on the high end of the types that don't disclose. To the buyer, her type can only be evaluated as best as $E(\theta \mid \text{undisclosed})$, which is strictly lower than her true type θ . In this case the seller is essentially being lumped in with the low θ 's that are also not disclosing, and that

is not good for her. So that means actually the seller should disclose. So then more types will disclose, and we can keep repeating this argument until all but the $\theta = 0$ will disclose, because as long as multiple types of θ are not disclosing then some θ in that group is being disadvantaged by the information asymmetry and the rest of the nondisclosing group.

Rather remarkably the mere introduction of a DMV completely reverses the market failure from no one willing to sell except the lemons to everyone willing to sell except the lemons.

This suggests that regulation, ie. a DMV, may solve the problem. However, the DMV probably won't give the seller θ for free because it probably costs them money to do an inspection etc. Or the DMV also wants to make a profit.

Suppose the DMV charges a fee of $c > 0$ for certificates. Then each seller chooses whether to buy a certificate. The certificate will display the car's true quality θ to the buyer. Sellers that don't buy the certificate will still have their θ concealed. However, an important implication of this is that the buyer is aware that some sellers on the market may disclose and others won't and can reason about this when evaluating a seller.

This means that the buyer is willing to pay for

- $w = \theta$ if θ is disclosed and
- $w = E(\theta \mid \text{undisclosed}) = \hat{\theta}$ if θ is not disclosed.

Given $\hat{\theta}$, the seller is better off disclosing if

$$\theta - c \geq \hat{\theta}$$

and better off not disclosing if

$$\theta - c < \hat{\theta}.$$

Thus,

$$\hat{\theta} = E(\theta \mid \text{undisclosed}) = E(\theta \mid \theta < \hat{\theta} + c) = \frac{\min(\hat{\theta} + c, 1)}{2},$$

which solves $2\hat{\theta} - \min(\hat{\theta} + c, 1) = 0$ ie. $\min(\hat{\theta} - c, 1 - 2\hat{\theta}) = 0$, which means $\hat{\theta} = \min(c, \frac{1}{2})$. If $c > \frac{1}{2}$, then $\hat{\theta} = \frac{1}{2}$ and $\theta - c < 1 - c < \frac{1}{2} = \hat{\theta}$, so no one will disclose. If $c \leq \frac{1}{2}$, then $\hat{\theta} = c$,

and the ones that disclose are $\theta \geq c + c = 2c$.

Given this, the DMV maximizes their profit of $P(c) := c(1 - 2c)$ for $c \leq \frac{1}{2}$. This is maximized at $c = \frac{1}{4}$, with DMV's profit being $\frac{1}{8}$ and only car dealers with $\theta \geq \frac{1}{2}$ disclose.

We see here the DMV is maximizing under a tradeoff. The DMV wants more sellers to buy the certificate, but by making the certificate's cost higher, fewer sellers will buy it. So by charging the fee for the certificate, the DMV is not able to ensure all types will disclose.